ILEGH Salma

Big Data & AI Engineering Student – Seeking a 4 to 6 Month Final-Year Internship (PFE)

Email: salmailegh@gmail.com | Phone: +212 623 790 795 | Casablanca, Morocco | github.com/salmailegh2004-lgtm

PROFILE

Big Data & AI engineering student at Université Internationale de Rabat (UIR), seeking a 4 to 6 month final-year internship (PFE) in Data and AI. Skills in data engineering, Big Data, Machine Learning and Deep Learning gained through academic projects and internships. Versatile and motivated, I aim to contribute to innovative projects, from data collection to turning data into value.

SKILLS

Programming: Python, SQL, Java, PHP

AI & Data Science: PyTorch, Scikit-learn, Pandas, NumPy, Machine Learning, Deep Learning, NLP

Big Data & Data Engineering: Hadoop (HDFS), Spark, Hive, Talend, NiFi, Data Warehouse, Azure, MongoDB, Neo4j

Generative AI & Tools: RAG, LLM, Llama 3, ChromaDB, FAISS, Ollama, FastAPI, Power BI

PROFESSIONAL EXPERIENCE

AI & Data Science Engineering Intern – RAG Chatbot | CNSS, Casablanca **June 2026**

Duration: 5 weeks | Python, FastAPI, ChromaDB, CrossEncoder, Ollama

- Designed a multilingual (French/Arabic) RAG chatbot based on a CrossEncoder reranker, ChromaDB vector storage and a local Ollama LLM.
- Built the document base by collecting and processing official documents (Playwright, pdfplumber).
- Developed the FastAPI backend with conversation logging in SQLite.

Project Observation Internship – Payment Platform | HPS, Casablanca **June 2025**

Duration: 5 weeks | Python, Scikit-learn, TF-IDF

- Analyzed card payment flows and handled incidents on the PowerCARD payment platform; modeled a machine learning classifier (TF-IDF, Scikit-learn) for bank fraud detection.

IT Observation Internship | BCP, Casablanca **June 2024**

Duration: 5 weeks | Data security, network monitoring

- Analyzed IT operational processes and data security principles (CIA triad) in a banking environment.

PROJECTS

Retail AI – Multi-Agent Analytics Platform **2025–2026**

PyTorch, LSTM, CrewAI, Streamlit, Ollama

- Developed a multi-agent platform for demand forecasting, anomaly detection and recommendation.
- Integrated LSTM and LSTM Autoencoder with agent orchestration and human validation; $R^2 = 0.56$ for forecasting.

Big Data Pipeline – Energy Consumption Forecasting **2025–2026**

Hadoop, Hive, Spark, Spark MLlib, Zeppelin

- Built a Big Data pipeline for the storage, processing and predictive analysis of energy consumption.
- Compared MLlib models; Random Forest achieved the best results (MAE 1.35; RMSE 1.75; R^2 0.33).

Cloud Data Warehouse – Oracle Decision-Support Platform **2025–2026**

Oracle SQL, Data Warehousing, Power BI

- Designed a distributed HR architecture based on horizontal and vertical fragmentation.
- Set up secure views, Materialized Views and synchronized Oracle indexes.

Hybrid Medical Triage – AI & Graph Database **2025–2026**

Ollama, FAISS, Neo4j, FastAPI

- Designed a hybrid system combining LLM, semantic search and graph-based reasoning for medical triage.
- Integrated a local LLM with Ollama, a FAISS vector database and a Neo4j knowledge graph.

QR & Barcode Detection System **2025–2026**

Python, OpenCV, NumPy, Pyzbar

- Detected and decoded QR codes and barcodes with OpenCV: preprocessing, contour detection, thresholding, Sobel, morphological operations.

EDUCATION

Engineering Degree in Big Data & AI | Université Internationale de Rabat (UIR) **2022–Present**

LANGUAGES

French (fluent) | **English** (professional) | **Arabic** (native)